

Group 2 Final Project: DataCo Supply Chain

Business Rules

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GBA 6220: Data Management Course

California State Polytechnic University, Pomona

Spring 2026

Overview

This document includes the business rules governing the DataCo Supply Chain relational database. The business rules define the constraints, relationships, and logic across all 11 entities. They cover entity integrity, referential integrity, cardinality, valid categorical values, nullable fields, and binary flag definitions. They serve as the foundation for the ER diagram and should be referenced by all team members throughout the database creation and data warehouse phases to ensure consistency across all deliverables.

Entity Summary

#	Entity	Description
1	CUSTOMER	Customers who place orders
2	DEPARTMENT	Store departments managing product categories
3	CATEGORY	Product categories grouping related products
4	PRODUCT	Products available in the supply chain catalog
5	SHIPPING_MODE	Available shipping methods for fulfilling orders
6	PAYMENT_TYPE	Accepted payment transaction types
7	MARKET	Global markets to which orders are delivered
8	ORDER_LOCATION	Delivery destination details for orders
9	ORDER	Customer orders placed through the system
10	LINE	Individual line items linking orders to products
11	SHIPMENT	Shipment and delivery details per order

DataCo Supply Chain Business Rules

1. CUSTOMER

- Each customer must have a unique CUSTOMER_ID
- Each customer must have a first name, last name, and segment
- Each customer must belong to exactly one segment (Consumer, Corporate, or Home Office)
- A customer may place one or many orders
- Customer address fields (CUSTOMER_STREET, CUSTOMER_CITY, CUSTOMER_STATE, CUSTOMER_COUNTRY) are required
- CUSTOMER_ZIPCODE is optional and nullable

2. DEPARTMENT

- Each department must have a unique DEPARTMENT_ID and a department name
- A department can be associated with one or many categories
- Every category must belong to exactly one department

3. CATEGORY

- Each category must have a unique PRODUCT_CATEGORY_ID and a category name
- Each category must belong to exactly one department
- A category can contain one or many products
- Every product must belong to exactly one category

4. PRODUCT

- Each product must have a unique PRODUCT_CARD_ID, a product name, and a product price
- Each product must belong to exactly one category
- PRODUCT_STATUS is a binary flag where 0 = available and 1 = not available

- A product can appear in one or many order lines across different orders

5. ORDER

- Each order must have a unique ORDER_ID
- Each order must be associated with exactly one customer via CUSTOMER_ID
- Each order must be associated with exactly one payment type via PAYMENT_TYPE_ID
- Each order must be associated with exactly one shipping mode via SHIPPING_MODE_ID
- Each order must be associated with exactly one delivery location via LOCATION_ID
- Each order must record an order date, order status, and order profit
- An order can have one of the following statuses: COMPLETE, PENDING, CLOSED, PENDING_PAYMENT, CANCELED, PROCESSING, SUSPECTED_FRAUD, ON_HOLD, or PAYMENT_REVIEW
- Each order must contain at least one order line
- Each order must have exactly one associated shipment

6. LINE (*Order Line Associative Entity/Table*)

- Each line must have a unique LINE_ID
- Each line must belong to exactly one order via ORDER_ID
- Each line must reference exactly one product via PRODUCT_CARD_ID
- Each line must record the following: quantity, discount amount, discount rate, product price, profit ratio, sales amount, and total amount
- LINE_SALES represents the gross sales value before discount is applied
- LINE_TOTAL represents the net amount after discount is applied
- LINE_TOTAL is always less than or equal to LINE_SALES
- An order must have at least one line
- A product can appear across many lines in different orders

7. SHIPMENT

- Each shipment must have a unique surrogate SHIPMENT_ID
- Each shipment must be associated with exactly one order via ORDER_ID
- Each shipment must record a shipping date, real shipping days, and scheduled shipping days
- Each shipment must have a delivery status of one of the following: Advance Shipping, Late Delivery, Shipping Canceled, or Shipping on Time
- LATE_DELIVERY_RISK is a binary flag where 0 = no risk and 1 = at risk
- SHIPPING_DATE must be stored as DATETIME
- If DAYS_FOR_SHIPPING_REAL exceeds DAYS_FOR_SHIPMENT_SCHEDULED, the shipment is considered late

8. SHIPPING_MODE

- Each shipping mode must have a unique surrogate SHIPPING_MODE_ID
- Each shipping mode must have a name
- Valid shipping modes are: First Class, Second Class, Standard Class, and Same Day
- A shipping mode can be assigned to one or many orders

9. PAYMENT_TYPE

- Each payment type must have a unique surrogate PAYMENT_TYPE_ID
- Each payment type must have a name
- Valid payment types are: Debit, Transfer, Cash, and Payment
- A payment type can be associated with one or many orders

10. MARKET

- Each market must have a unique surrogate MARKET_ID
- Each market must have a name

- Valid markets are: Africa, Europe, LATAM, Pacific Asia, and USCA
- A market can contain one or many order locations
- Every order location must belong to exactly one market

11. ORDER_LOCATION

- Each location must have a unique surrogate LOCATION_ID
- Each location must belong to exactly one market via MARKET_ID
- Each location must record a city, state, country, and region
- ORDER_ZIPCODE is optional and nullable
- A location can be associated with one or many orders
- Each order must be delivered to exactly one location